

Key Resume Attributes Impacting Job Callbacks

Stanford STATS209 Final Project

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1. Project description

Motivation and Problem statements. As we enter the labor market ourselves, we are curious about how different credentials, race, and gender can impact our chances of getting callbacks. In the early 2000s, researchers wondered the same, but they approached the problem with simple linear regressions and somewhat opaque methods for "controlling for" covariates. We want to revisit the data with approaches learned in class, to see how the results hold up today.

Data. The dataset originates from "Key Resume Attributes Impacting Job Callbacks" on Kaggle, exploring the impact of race and gender on job application callback rates in Boston and Chicago during 2001 and 2002. Job advertisements are identified by job_ad_id and include location, industry, type, job requirements, and other details.

Applicant details cover first names associated with race and gender (such as Emily for white female and Jamal for African-American male), education, work-related skills, and indicators for honors, volunteer experience, military service, and employment gaps. Resumes are categorized as lower or higher quality, with received_callback indicating callback status. This dataset is valuable for analyzing factors affecting callback rates in the labor market.

Related work. In the realm of labor market discrimination, a pivotal study conducted by Bertrand and Mullainathan (2004) stands out as a landmark investigation. The study utilized fictitious resumes, varying only in the ethnic or racial associations of the names (e.g., Emily and Greg as perceived white candidates, and Lakisha and Jamal as perceived African-American candidates), and submitted them in response to real-world job advertisements. The findings of this research provided evidence of significant racial disparities in employer callback rates, shedding light on the persistence of discriminatory practices in the labor market. The study's methodological approach and results serve as a foundational

reference, prompting further exploration into the study of labor market discrimination.

2. Methods and Experiments.

In the original study, resumes were sent in response to help-wanted ads in Chicago and Boston newspapers, and the manipulation of gender and race perception was achieved by **randomly assigning** either very White-sounding names for women and men (e.g., Emily Walsh or Greg Baker) or very African-American-sounding names for women and men (e.g., Lakisha Washington or Jamal Jones) to the resumes. The key element defining this experiment as completely randomized is the random assignment of names to resumes. With randomization, the experiment ensures that any observed differences in callback rates for interviews can be attributed to the manipulation of gender and race perception rather than any pre-existing characteristics of the applicants. This random assignment helps control for potential confounding variables, allowing for a more reliable assessment of the causal impact of discrimination on callback rates in the labor market. Therefore, we can treat this experiment as a completely randomized experiment.

For the analysis of the impact of race and gender on job application callbacks using the dataset, we propose employing Fisher's Randomization Test (FRT) and Neyman's Causal Inference. FRT will be implemented using a permutation-based approach, where treatment assignments are shuffled to create a distribution under the null hypothesis. Neyman's framework will be used to estimate the average treatment effect of race/gender indicators on callback rates. This involves comparing average callback rates between different treatment and control groups.

To bolster the robustness of our analysis, we have incorporated stratified approaches for gender, race, and different industries. Additionally, we've employed matched pairs, considering gender, race, or resume quality as the treatment. This strategic pairing ensures a more nuanced exploration of the interplay between these variables, providing a deeper

understanding of how race and gender dynamics intersect with callback rates in diverse contexts.

In the subsequent sections, we delve into the specific findings and implications derived from the application of different methods, shedding light on the multifaceted relationships between race, gender, and job application outcomes.

2.1. FRT

In our study, we first use Fisher’s Randomization Tests (FRT) to understand how race and gender affect callback rates. Based on randomization test, we assume a sharp null hypothesis of no effect and then compute the exact p-value, that is, the probability of seeing an effect as big as the one observed assuming no effect.

FRT for gender We are interested in testing the following null hypothesis: $H_{0F} : Y_i(1) = Y_i(0)$ for all units $i = 1, \dots, n$ with the treatment $Z_i \in \{0(\text{female}), 1(\text{male})\}$. We use difference-in-means as our test statistic here. Under the null hypothesis of no difference between groups, we obtain the p-values of 0.379, which is relatively high. This indicates a lack of significant evidence to reject the null hypothesis in favor of a difference between groups.

FRT for race We evaluate the null hypothesis $H_{0R} : Y_i(1) = Y_i(0)$ for all units $i = 1, \dots, n$, where the treatment variable $Z_i \in \{0(\text{African American}), 1(\text{White})\}$. We adopt difference-in-means as our test statistics.

In initial 10,000 permutations, the test statistics yielded p-values of 0. We were expecting some causal effects but we weren’t expecting it to be this strong. This result implies that none of the simulated values of the test statistics, representing differences in callback rates, were as extreme as the observed value. Consequently, strong evidence emerges against the null hypothesis.

To explore further, we increased the number of permutations to 1,000,000. Despite this increase, the p-value is $3.5e-05$, which remains remarkably small. These persistently low p-values reinforce the conclusion that the observed data significantly deviates from what would be expected under the null hypothesis, providing robust evidence against the equality of callback rates for African American and White individuals in our study.

FRT combining race and gender In our pursuit of a deeper understanding, we seek to examine the potential impact of different combinations of races and genders on callback rates. To accomplish this, we partition the data into four distinct categories defined by the combinations of race and gender: $(Z_r, Z_g) \in \{(1, 1), (1, 0), (0, 1), (0, 0)\}$, where Z_r denotes race (1 for White, 0 for African American), and Z_g represents gender (1 for Male, 0 for Female). Having

categorized the data into the four combinations, we proceed to conduct group-wise FRT among these subgroups using difference-in-means as the test statistics to get the effect between each two groups.

Treated group	Control group	P-value
White Male (1, 1)	White Female (1, 0)	0.4366
White Male (1, 1)	Black Male (0, 1)	0.0628
White Male (1, 1)	Black Female (0, 0)	0.0836
White Female (1, 0)	Black Male (0, 1)	0.0018
White Female (1, 0)	Black Female (0, 0)	0.0002
Black Male (0, 1)	Black Female (0, 0)	0.5432

Table 1. P-values for Different Test Statistics

From the table, it is evident that two out of the six p-values are less than the significance level of 0.05, rejecting the null hypothesis of no causal effect between race/gender and callback rate. Notably, both of the two p-values rejecting null hypothesis are associated with comparisons among candidates of different racial backgrounds, comparing White women with African American candidates. Although the other two p-values associated with comparisons among White men and African American candidates are not lower than the significance level of 0.05, both of the p-values are close to 0.05, such that even though we are not rejecting the null, we should be cautious and recognize the marginal nature of the results. This indicates statistically significant differences in callback rates based on race, emphasizing the importance of considering race in the context of labor market discrimination.

It’s interesting to observe that when comparing white candidates and African American candidates, white females appear to have a lower p-value than white males. From Table 1, it is evident that, in the context of comparing with African American candidates, the p-values for white females are lower than those for white males. This suggests a potential difference in callback rates that favors white females over white males.

2.2. Neymanian Repeated Sampling Inference

We also adopt Neyman’s methodology, that involves estimating the average treatment effect and a conservative variance. Based on the observed outcomes, we can calculate the sample means and compute the unbiased difference-in-means estimator $\hat{\tau}$. Then, we compute the conservative variance estimator for estimating $var(\hat{\tau})$ and construct a

95% confidence interval.

$$\begin{aligned}\hat{Y}(1) &= n_1^{-1} \sum_{i=1}^n Z_i Y_i, \quad \hat{Y}(0) = n_0^{-1} \sum_{i=1}^n (1 - Z_i) Y_i \\ \hat{\tau} &= \hat{Y}(1) - \hat{Y}(0) \\ \hat{V} &= \frac{\hat{S}^2(1)}{n_1} + \frac{\hat{S}^2(0)}{n_0} \\ \hat{se} &= \sqrt{\hat{V}} = \text{estimated standard error}\end{aligned}$$

Leveraging the Central Limit Theorem (CLT), this framework empowers us to make meaningful inferences by relying on a normal approximation.

Neymanian inference for gender The result on Table 2 suggests that based on the Neymanian inference, we are 95% confident that the true average treatment effect for gender falls within the range of -0.0263 to 0.0090. The inclusion of zero in the confidence interval implies that there is a possibility of no treatment effect. In other words, we cannot rule out the possibility that the observed differences are due to random variation rather than a systematic treatment effect.

$\hat{\tau}$	$\hat{se}$	95% confidence interval
-0.00864	0.0090	[-0.0263 , 0.0090]

Table 2. $\hat{\tau}$, estimated std. error, and confidence interval with Neyman's method

To further explore, we use regression-based inference for the average causal effect τ by running the ordinary least squares of the outcomes on the treatment indicators with an intercept and use the coefficient of the treatment $\hat{\beta}$ as the estimator for the average causal effect.

$$(\hat{\alpha}, \hat{\beta}) = \arg \min_{(a,b)} \sum_{i=1}^n (Y_i - a - bZ_i)^2$$

$\hat{\tau}$	$\hat{se}$	95% confidence interval
-0.00864	0.0093	[-0.02678 , 0.009492]

Table 3. $\hat{\tau}$, estimated std. error, and confidence interval using OLS

The above standard error is very close to the one in Table 2 based on the formulas stated in 2.2 but they are not the same. We further explore with EHW robust standard error, and can get a standard error of 0.0090 that is even closer to the estimated standard error in Table 2.

Neymanian inference for race With the treatment being the race of the candidate, the result on the Table 4 gives a 95% confidence interval between 0.0168 and 0.0473. Since zero is not included in the interval, we are 95% sure that there is a treatment effect on the callback rate.

$\hat{\tau}$	$\hat{se}$	95% confidence interval
0.0320	0.007785	[0.0168 , 0.0473]

Table 4. $\hat{\tau}$, estimated std. error, and confidence interval with Neyman's method

We further implement the regression-based inference for τ using OLS and got the results in Table 5. The results are the same as using the formulas in 2.2.

$\hat{\tau}$	$\hat{se}$	95% confidence interval
0.0320	0.007785	[0.0168 , 0.0473]

Table 5. $\hat{\tau}$, estimated std. error, and confidence interval using OLS

2.3. FRT in Industry Stratification

The examination of callback rates in the job market, when stratified by gender and race across different industries, unveils significant variations and patterns that might be obscured in a more generalized analysis.

Industry	Female Call-back Rate	Male Call-back Rate
Business and personal service	8.16%	9.33%
Finance, insurance, and real estate	7.20%	7.69%
Manufacturing	6.69%	2.50%
Other Service	9.17%	7.56%
Transportation and communication	13.19%	14.04%
Wholesale and retail trade	7.04%	6.74%

Table 6. Proportion of Callback Rates by Industry with Treatment Gender

We observe notable differences in callback rates between female and male candidates across industries as the Table 6 showed. For instance, in the "manufacturing" industry, the callback rate for females is 6.69%, compared to 2.50% for males. Such disparities suggest potential gender biases or differential impacts of gender across industries.

Similarly, when we focus on race, the proportion of callback rate of different races in Table 7, there are even more pronounced disparities. In industries like "Finance, Insurance, and Real Estate", the callback rate for White candidates is substantially higher (10.14%) compared to Black candidates (4.38%). This stark contrast highlights the racial disparities in the job market, suggesting systemic biases or discrimination against certain racial groups in specific sectors.

Industry	White People Callback Rate	Black People Callback Rate
Business and personal service	10.43%	6.29%
Finance, insurance, and real estate	10.14%	4.38%
Manufacturing	6.93%	3.96%
Other Service	10.01%	7.83%
Transportation and communication	12.16%	14.86%
Wholesale and retail trade	8.64%	5.18%

Table 7. Proportion of Callback Rates by Industry with Treatment Race

Stratified FRT for gender The method here is conducting Fisher’s Randomization test in every stratum. We have a similar null hypothesis as in 2.1, which is $H_{0F} : Y_i(1) = Y_i(0)$ for each stratum k with $i = 1, \dots, n_k$ with the treatment $Z_i \in \{0 \text{ (female)}, 1 \text{ (male)}\}$. We use different-in-means as our test statistics and 10,000 permutations as before. Based on the result of Table 8, none of the test statistics is significant (all p-values > 0.05). We do not have enough evidence to reject the null. Thus we can conclude that the callback rates have no difference between genders in all industries.

Industry	FRT p-value
Business and personal service	0.5995
Finance, insurance, and real estate	1
Manufacturing	0.1022
Other Service	0.4572
Transportation and communication	1
Wholesale and retail trade	0.9014

Table 8. FRT p-value by industry with treatment gender

Stratified FRT for race In this section, we evaluate the null hypothesis $H_{0F} : Y_i(1) = Y_i(0)$ for each stratum k with $i = 1, \dots, n_k$ with the treatment $Z_i \in \{0 \text{ (African American)}, 1 \text{ (White)}\}$ using difference-in-means as test statistics and 10,000 permutations performed for each industry. In industries of "business and personal service" (p-value = 0.0074), "finance, insurance, and real estate" (p-value = 0.0386) and "wholesale and retail trade" (p-value = 0.0351) from Table 9, we have enough evidence to reject the null and can conclude that there is difference between races in callback rates among these three industries. In the other three industries, we can not reject the null due to p-values are greater than 0.05. This suggests no noticeable racial disparity in callback rates in manufacturing, other services and transportation and communication.

The analysis reveals a notable trend: industries that demand advanced technological skills or high professionalization exhibit significant racial disparities in callback rates. This is particularly evident in sectors like "finance, insurance, and real estate". These findings suggest the presence of

biases or systemic barriers in these specialized fields, highlighting the need for focused efforts to address racial equity in areas requiring higher technical expertise or professional qualifications

Industry	FRT p-value
Business and personal service	0.0074
Finance, insurance, and real estate	0.0386
Manufacturing	0.2672
Other Service	0.1569
Transportation and communication	0.8059
Wholesale and retail trade	0.0351

Table 9. FRT p-value by industry with treatment race

2.4. Neymanian Repeated Sampling Inference Across Industry Strata

Stratified Neymanian inference for gender From Table 10, we notice that 95% confidence intervals for all industries except manufacturing include zero. This means we are 95% confident that all industries except manufacturing have no significant difference in callback rates between genders. $\hat{\tau} = 0.041901$ for manufacturing suggests that male has a higher average callback rate compared to female in the manufacturing industry.

Industry	$\hat{\tau}$	$\hat{se}$	95% CI
Business and personal service	-0.011776	0.000447	[-0.053228, 0.029675]
Finance, insurance, and real estate	-0.004923	0.002047	[-0.093606, 0.083760]
Manufacturing	0.041901	0.000425	[0.000425, 0.082328]
Other Service	0.016036	0.000359	[-0.021054, 0.053126]
Transportation and communication	-0.008483	0.003426	[-0.123214, 0.106248]
Wholesale and retail trade	0.002936	0.000251	[-0.028137, 0.034009]

Table 10. $\hat{\tau}$, $\hat{se}$, and confidence interval for each industry, gender treatment

Stratified Neymanian inference for race Table 11 shows in "business and personal service", "finance, insurance, and real estate" and "wholesale and retail trade" industries, the 95% confidence intervals do not include zero, and $\hat{\tau} < 0$. The 95% confidence intervals of the rest industries ("manufacturing", "other Service" and "transportation and communication") include zero. Based on Neymanian inference, significant racial disparities favoring white individuals are observed in which industries the confidence interval does not include zero. These results meet the earlier conclusion in Section 2.3.

Industry	$\hat{\tau}$	$\hat{se}$	95% CI
Business and personal service	-0.041411	0.000234	[-0.071394, -0.011428]
Finance, insurance, and real estate	-0.057971	0.000644	[-0.107725, -0.008217]
Manufacturing	-0.029703	0.000510	[-0.073972, 0.014566]
Other Service	-0.021823	0.000209	[-0.050130, 0.006484]
Transportation and communication	0.027027	0.003197	[-0.083796, 0.137850]
Wholesale and retail trade	-0.034549	0.000246	[-0.065306, -0.003792]

Table 11. $\hat{\tau}$, estimated std. error, and confidence interval for each industry controlling race

2.5. Matched Pairing

Matched pairing is used to re-examine the data through yet another lens. Since there are over 30 variables associated with each candidate's job application, propensity score calculations can be useful for matching the most similar candidates. Some variables were transformed to make the propensity score calculations possible (details in Github Repo). In this analysis, we explore race, gender, and resume quality as separate treatments.

Race as a "Treatment" Most variables were evenly balanced across race. This is reflected in the racial propensity score (mean 0.500, min 0.404, max 0.623). Indeed, the variables are so well randomized across race that attempts to add propensity calipers or exact gender matches resulted in essentially no change to matched pairs (vast majority of variables have ± 0.02 standardized differences, all < 0.06). The results from this analysis, shown in Table 12, are similar to what we previously found.

$\hat{\tau}$	$\hat{V}$	FRT p-value
-0.0305	0.000058	0.0000

Table 12. $\hat{\tau}$, estimated variance, and FRT p-value for Matched Pairing with Black assigned as Treatment, 2435 pairs

Gender as a "Treatment" Unlike for race, the covariate balance for gender was lacking. Gender was probably not randomly assigned to the resumes as the data publishers claim (gendered propensity score: mean 0.769, min 0.292, max 0.997). Before matching, the standardized differences were larger by more than an order of magnitude, with some larger than 1, as can be seen in Figure 1. The propensity score caliper made the most positive adjustment to balance, and although exact matching for race made slightly worse

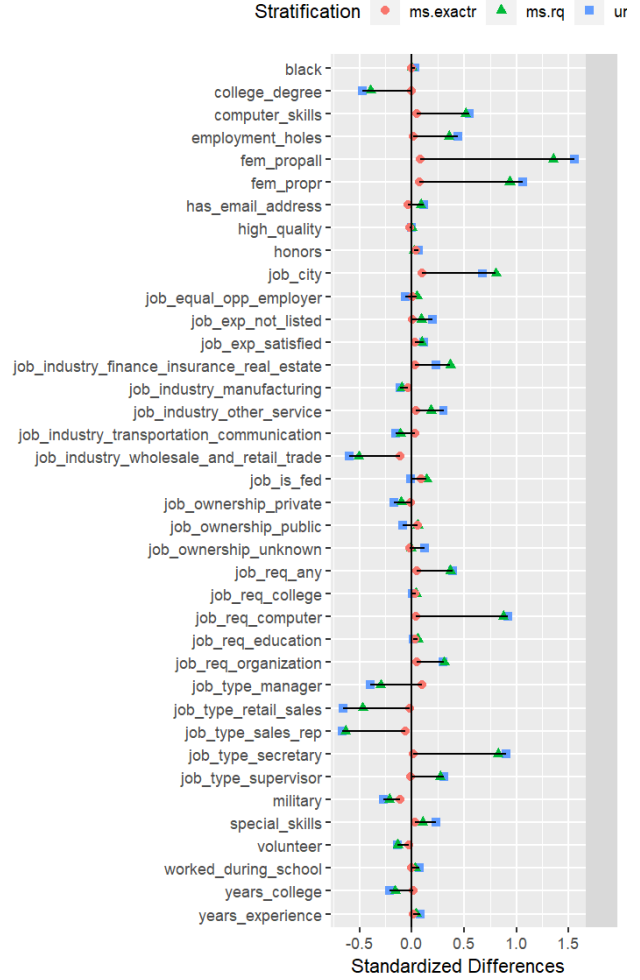


Figure 1. Gendered Covariate Balance across 3 sets of matched pairs (unstrat=unstratified, ms.rq= matched on race and resume quality, ms.exactr= ms.rq with gendered propensity caliper and exact race matching)

matches, we chose to do so anyway out of a desire to only capture gender causal effects. Table 13, are similar to what we previously found, using the matched set on quality and exact race, with a gendered propensity score caliper.

$\hat{\tau}$	$\hat{V}$	FRT p-value
-0.00544	0.000115	0.466

Table 13. $\hat{\tau}$, estimated variance, and FRT p-value for Matched Pairing with Female assigned as Treatment, 1072 pairs

The causal effect of gender is next to nothing and insignificant with its very high p-value. However, this warrants discussion because it does not mean that gender would have no effect on callbacks in the real world. Job type was one of

the most outstandingly unbalanced covariates. Bertrand and Mullainathan mention the fact that they "use male and female names for sales jobs, whereas [they] use nearly exclusively female names for administrative and clerical jobs to increase callback rates" (Bertrand and Mullainathan (2004), pg 996). Figure 2 shows that the dataset has many more women than men (3746 vs 1124) in addition to the job type imbalance. When more women apply to more traditionally feminine jobs, where they may be taken more seriously (Doering and Thébaud (2017)), this may obscure potential sexism by giving the women more and better chances to get callbacks. Thus, this result could just mean that when women (or men) apply to traditionally feminine jobs, their gender does not make a difference for whether they get a callback or not.

Next, we subset the data to remove the "secretary" and "clerical" roles which are heavily dominated by women. Once again, the propensity score caliper improved the balance most, and we used exact matching on race. Here, in Table 14, the gendered effect became slightly more significant but reduced in magnitude, even closer to no effect. Adding Job Ad ID to the matching conditions to make the job type comparison more granular resulted in worse matches and even weaker support for gender differences (FRT p-value= 0.737).

$\hat{\tau}$	$\hat{V}$	FRT p-value
-0.00338	0.000120	0.443

Table 14. $\hat{\tau}$, estimated variance, and FRT p-value for Matched Pairing on Non-Female-Dominant Subset, with Female assigned as Treatment, 1072 pairs

Without further details about the dominant gender of the specific roles, we cannot use this data to draw conclusions about gender differences in callbacks. For example, clerks in sales are male-dominated whereas clerks in office or administrative support are female-dominated (38% vs 88% women) (Bureau (2019)). The study would have more convincing results if it had sent applications in an organized mix of gendered job types. Then we could have seen if gender differences in callbacks exist for male-dominant and gender-neutral jobs. All of our results only apply to the types of jobs in this data set, ignoring important occupational segregation effects, by both gender and race.

Resume Quality as a "Treatment" Bertrand and Mullainathan had also assigned a binary quality of resume (high or low) to investigate how much resume quality pays off. They found that higher quality resumes did have a significant positive effect on callbacks, but the benefit was much larger for white applicants than black applicants. We were curious about potential gender intersectionality effects that they did not investigate, so we performed matched pairing

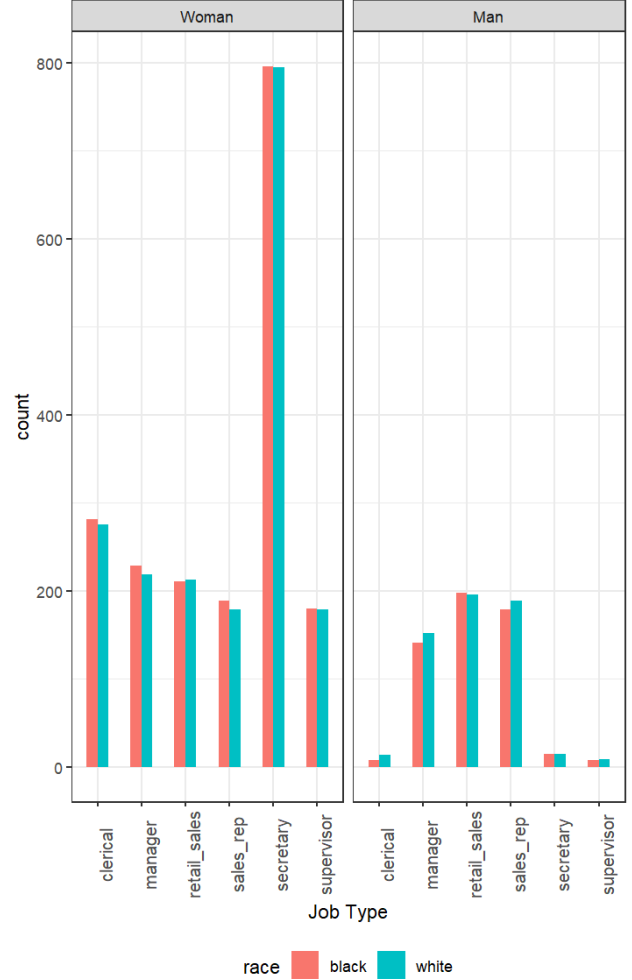


Figure 2. Unbalanced and Biased Job Application Choices. Race is clearly balanced while gender is not, even though both traits were assigned by researchers. Many female applicants were funneled to clerical and secretary jobs which are >93% dominated by women, according to the US Census Bureau's Detailed Occupation by Sex Education Age Earnings: ACS 2019.

on multiple subsets of data. First, we did the overall population, matched by race and gender, then we split across the race binary to simulate the original paper. Next, we split across the gender binary. Finally, we split into four dual-identity groups (White Man, White Woman, Black Man, Black Woman). Each subgroup had its own matched pairings, matched by at least two variables. When matching a subset by a demographic trait was not possible, they were matched by job application location (Chicago/Boston) and years of experience listed on resume. Results are displayed in Table 15 and Figure 3.

Higher resume quality seems to benefit white women the most and black men the least. We find that resume quality

Subset	$\hat{\tau}$	$\hat{V}$	FRT p-value
All	0.0404	0.000059	0.0306
White	0.0526	0.000135	0.0319
Black	0.0524	0.000095	0.309
Man	-0.0163	0.000219	0.575
White Man	0.178	0.000458	0.198
Black Man	0.325	0.000344	0.942
Woman	0.0812	0.000077	0.0171
Black Woman	0.0479	0.000124	0.0867
White Woman	0.131	0.000180	0.0508

Table 15. $\hat{\tau}$, $\hat{V}$, and FRT p-value for each matched pairs subset, with Resume Quality as Treatment

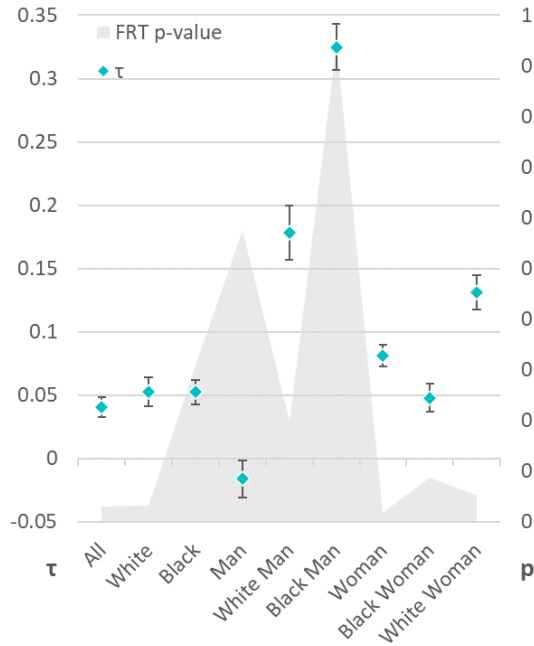


Figure 3. Visual Representation of Matched Pairing Results from Table 15. Lower p-values for white people and women suggest that both gender and race play a role in resume quality payoff.

most likely does not make a difference for callbacks for men, although it might matter more for white men. With $p > 0.94$, there is very likely no difference in callbacks for black men with varying quality resumes, but this is to be contrasted with black women $p < 0.09$ who likely benefit from a better quality resume. Indeed, black women may benefit more than the overall population, but the average effect for them is less than half of that of the average effect for white women. Our findings align with the original paper's conclusion that "employers simply seem to pay less attention or discount

more" black applicants' resume credentials (Bertrand and Mullainathan (2004), pg 1002). But we also find a significant gender difference, that women's resumes face more scrutiny than men's resumes in a way that slightly benefits black women with credentials.

These findings align with some ideas in literature. For example, affirmative action, which aims to increase credentials for minorities, is known to have most benefited white women (for example, see Wise (1998)). In "A Theory of Racialized Organizations", partially inspired by Bertrand and Mullainathan, Ray describes whiteness as a credential and blackness as a negative credential (a criminal record is another example of a negative credential). Credentials on resumes are supposed to be a "race-neutral" decision-making tool for hiring the "most qualified" candidates, so a racist narrative is reinforced when more white people are hired at an institution, even though the likely cause is hiring managers' implicit bias (Ray (2019)). Accordingly, standardized hiring procedures have been found to benefit women (Doring and Thébaud (2017), pg 564) while simultaneously institutionalizing racism. On the other hand, the extra resume quality benefits we see for women could stem from the female-dominated job types in this data set.

3. Conclusion.

Bertrand and Mullainathan (2004) found that black applicants had to submit about 50% more applications than white applicants to get a callback and that better credentials did not help black applicants nearly as much as they helped white applicants. They also found a "statistically insignificant" "reverse gender gap." We expected to find stronger evidence of racism and sexism, as well as more insights into the resume quality effects.

We found strong evidence to support the original conclusion that racial discrimination exists in the resume review process. We also confirmed the lack of gender differences in callbacks. But we want to emphasize that this could be due to the unbalanced distribution of resumes to job types and industries that are female-dominated. From this data, it is impossible to discern important effects of occupational segregation by gender and race.

We did find gender differences that were overlooked by Bertrand and Mullainathan. Higher quality resumes disproportionately benefit white women and may not cause any difference for men, especially black men. Racial disadvantages persist across gender, but black women with higher quality resumes seem to benefit from being women, with regards to their credentials being recognized by potential employers.

Code availability. The code for the data processing and analyses conducted above can be found at the following GitHub repository: Github Repo.

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